Year 12 Maths - Trigonometric functions and equations

Similar questions - Proving and using identities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Prove that $\tan\theta + \dfrac{1}{\tan\theta} \equiv \dfrac{1}{\sin\theta\cos\theta}$ $(\sin\theta\cos\theta \neq 0)$. **[3]**

(b) Solve $\tan x + \dfrac{1}{\tan x} = \dfrac{5}{2}$ for $0^\circ < x < 90^\circ$, giving your answers to 1 decimal place. **[4]**

(c) Explain why $\tan x + \dfrac{1}{\tan x} = 1$ has no solutions for $0^\circ < x < 90^\circ$. **[2]**

2

(a) Prove that $\dfrac{\sin x}{1+\cos x} + \dfrac{1+\cos x}{\sin x} \equiv \dfrac{2}{\sin x}$ $(\sin x \neq 0)$. **[4]**

(b) Hence solve $\dfrac{\sin x}{1+\cos x} + \dfrac{1+\cos x}{\sin x} = 4$ for $0^\circ < x < 360^\circ$. **[3]**

Question 1

(a) $\tan\theta + \dfrac{1}{\tan\theta} = \dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\sin\theta} = \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \dfrac{1}{\sin\theta\cos\theta}$

(b) Let $t = \tan x$: $t + \frac{1}{t} = \frac{5}{2}$, so $2t^2 - 5t + 2 = 0$ and $(2t-1)(t-2) = 0$. $\tan x = 2$ gives $x = 63.4°$; $\tan x = \frac{1}{2}$ gives $x = 26.6°$

(c) $t + \frac{1}{t} = 1$ gives $t^2 - t + 1 = 0$ with discriminant $1 - 4 = -3 < 0$, so there is no real t. (Alternatively, $\sin\theta\cos\theta \leqslant \frac{1}{2}$, so $\frac{1}{\sin\theta\cos\theta} \geqslant 2$.)

Question 2

(a) Common denominator: $\dfrac{\sin^2 x + (1+\cos x)^2}{\sin x(1+\cos x)} = \dfrac{\sin^2 x + 1 + 2\cos x + \cos^2 x}{\sin x(1+\cos x)} = \dfrac{2 + 2\cos x}{\sin x(1+\cos x)} = \dfrac{2}{\sin x}$

(b) $\dfrac{2}{\sin x} = 4$, so $\sin x = \frac{1}{2}$ and $x = 30°,\ 150°$